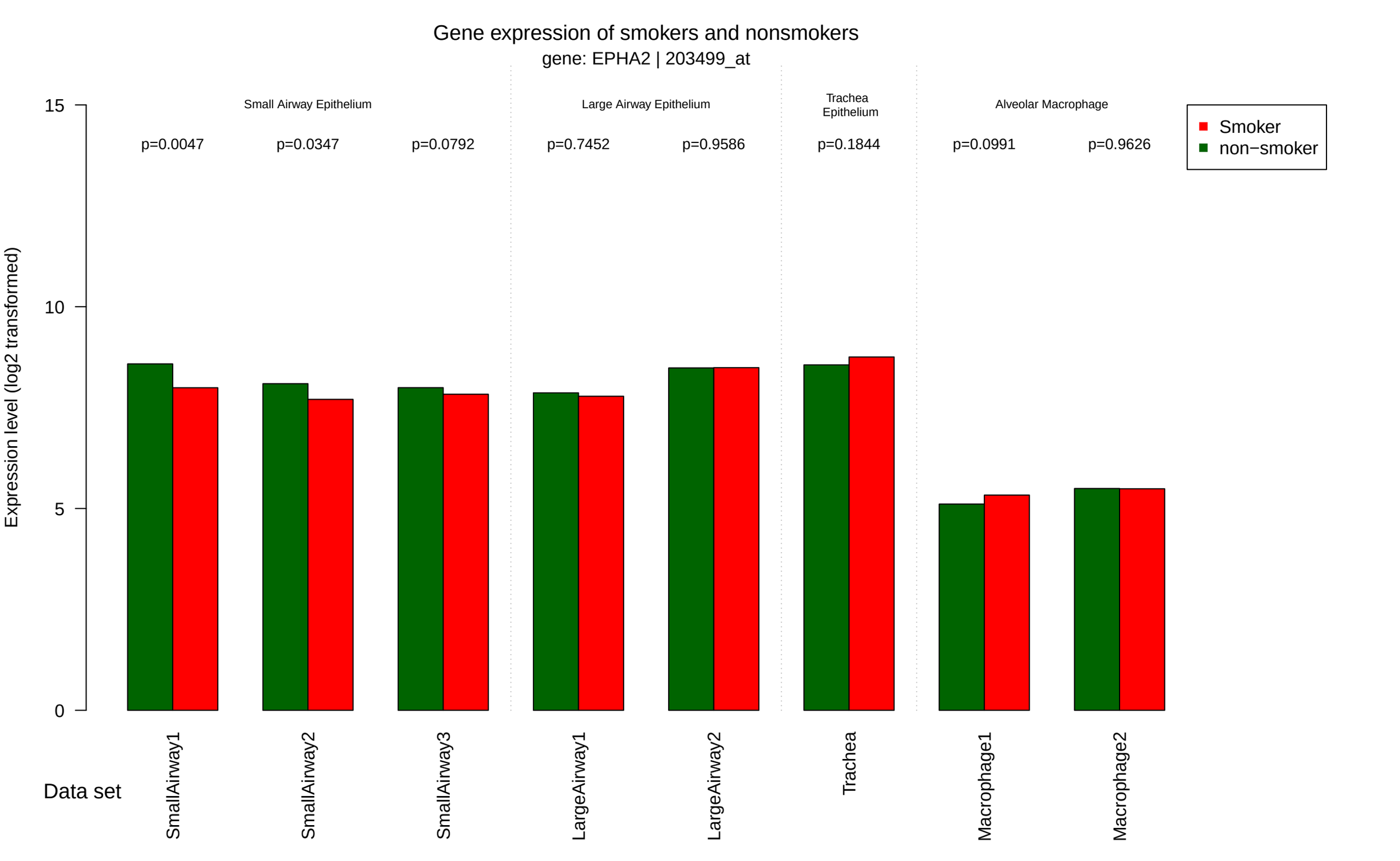
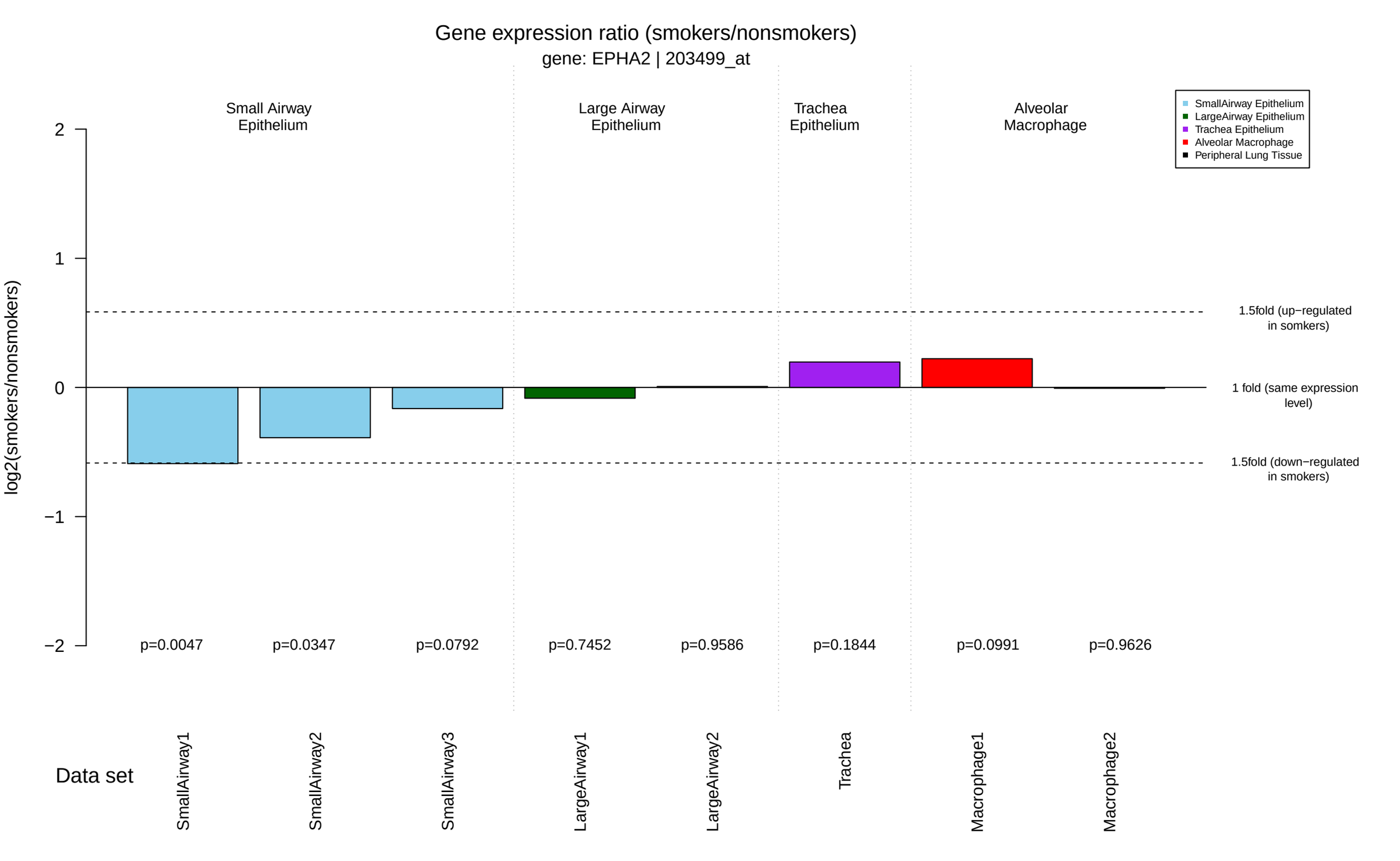
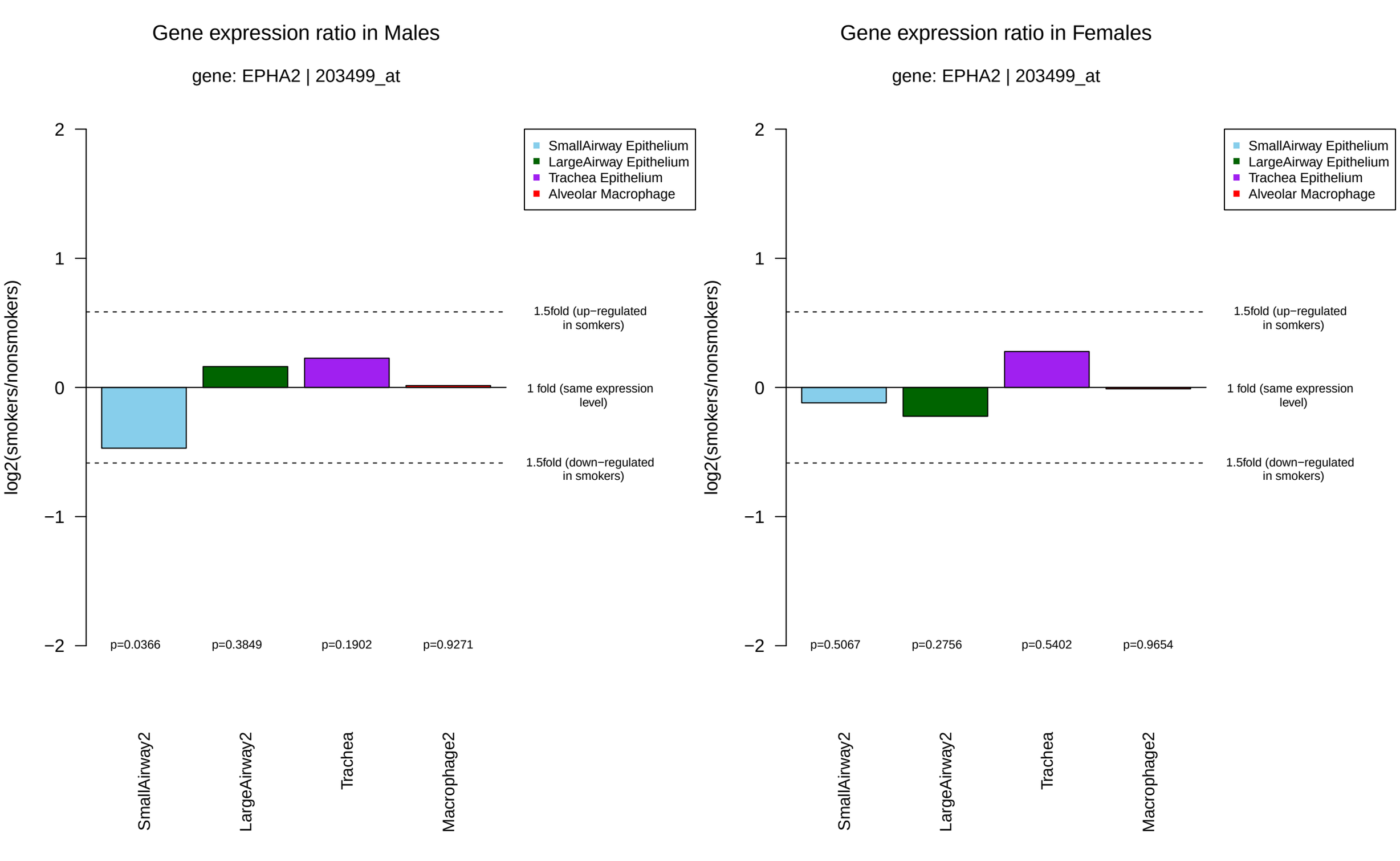
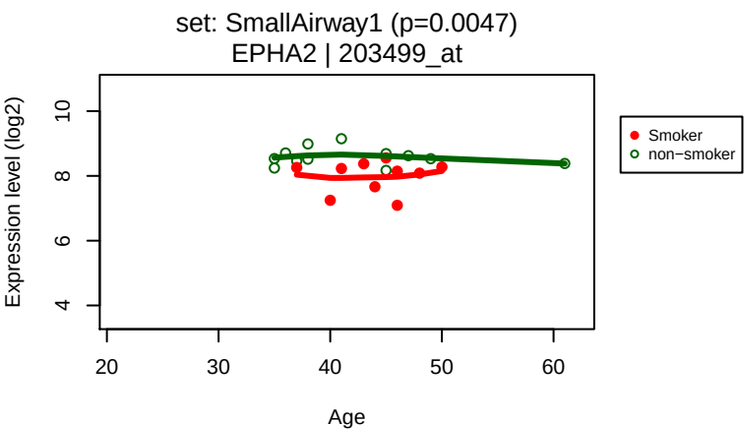
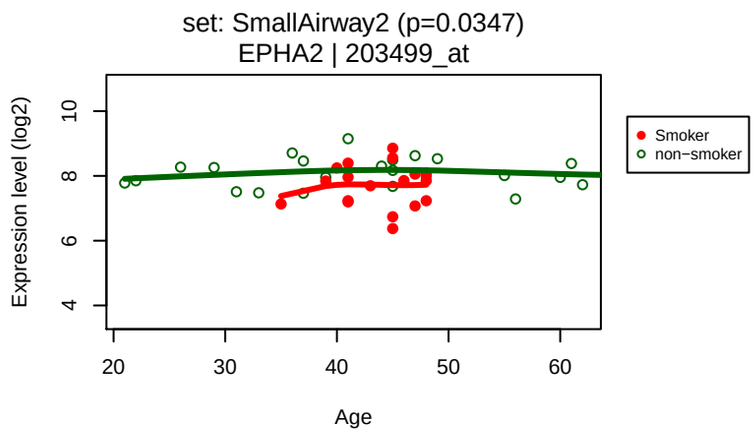
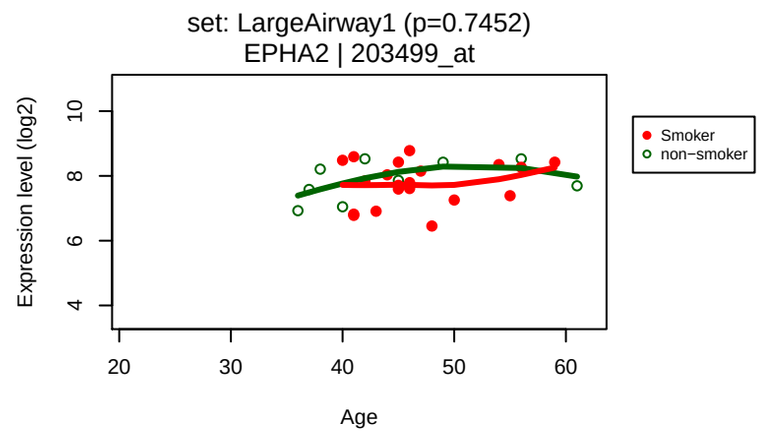

Gene expression of smokers and nonsmokers
gene: EPHA2 | 203499_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0047
p=0.0347
p=0.0792
p=0.7452
p=0.9586
p=0.1844
p=0.0991
p=0.9626
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

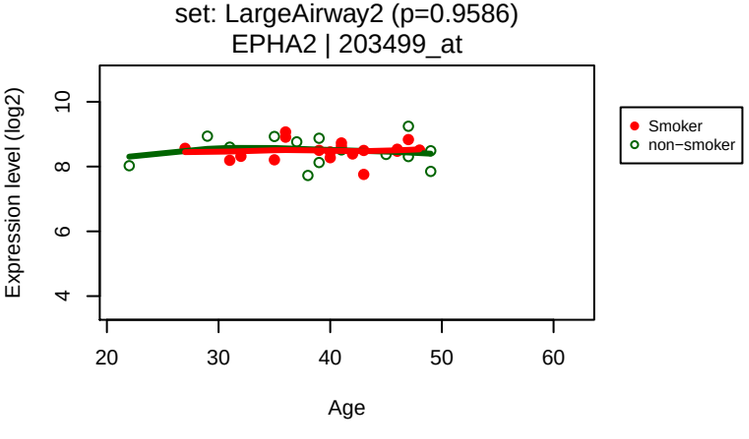

Gene expression ratio (smokers/nonsmokers)
gene: EPHA2 | 203499_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0047
p=0.0347
p=0.0792
p=0.7452
p=0.9586
p=0.1844
p=0.0991
p=0.9626
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

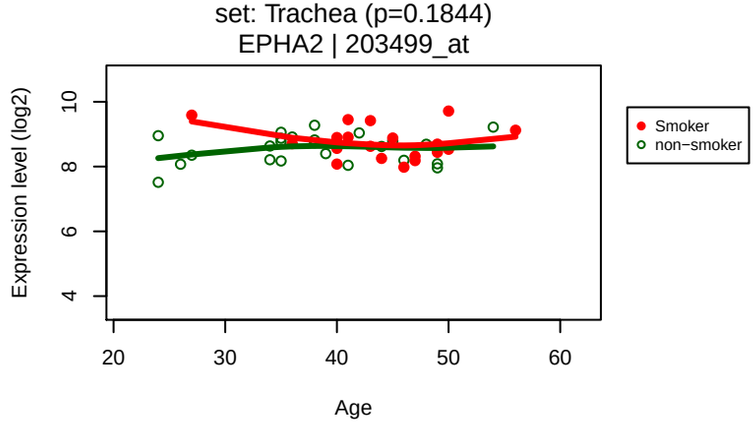

Gene expression ratio in Males
gene: EPHA2 | 203499_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0366
p=0.3849
p=0.1902
p=0.9271
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EPHA2 | 203499_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5067
p=0.2756
p=0.5402
p=0.9654
SmallAirway2
LargeAirway2
Trachea
Macrophage2

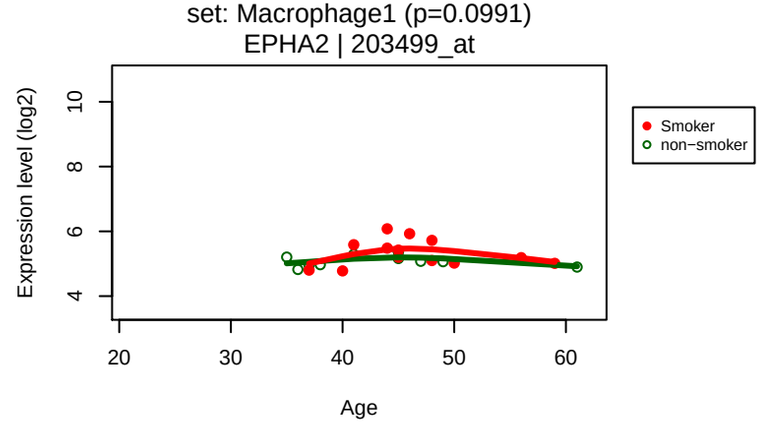

set: SmallAirway1 (p=0.0047)
EPHA2 | 203499_at
Expression level (log2)
Age
Smoker
non−smoker

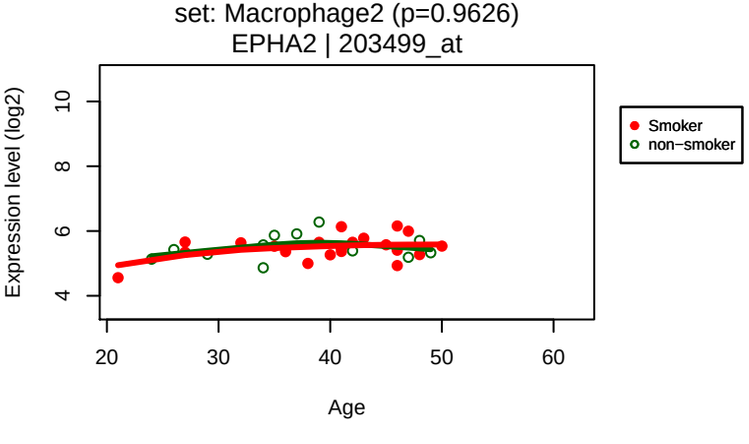

set: SmallAirway2 (p=0.0347)
EPHA2 | 203499_at
Expression level (log2)
Age
Smoker
non−smoker

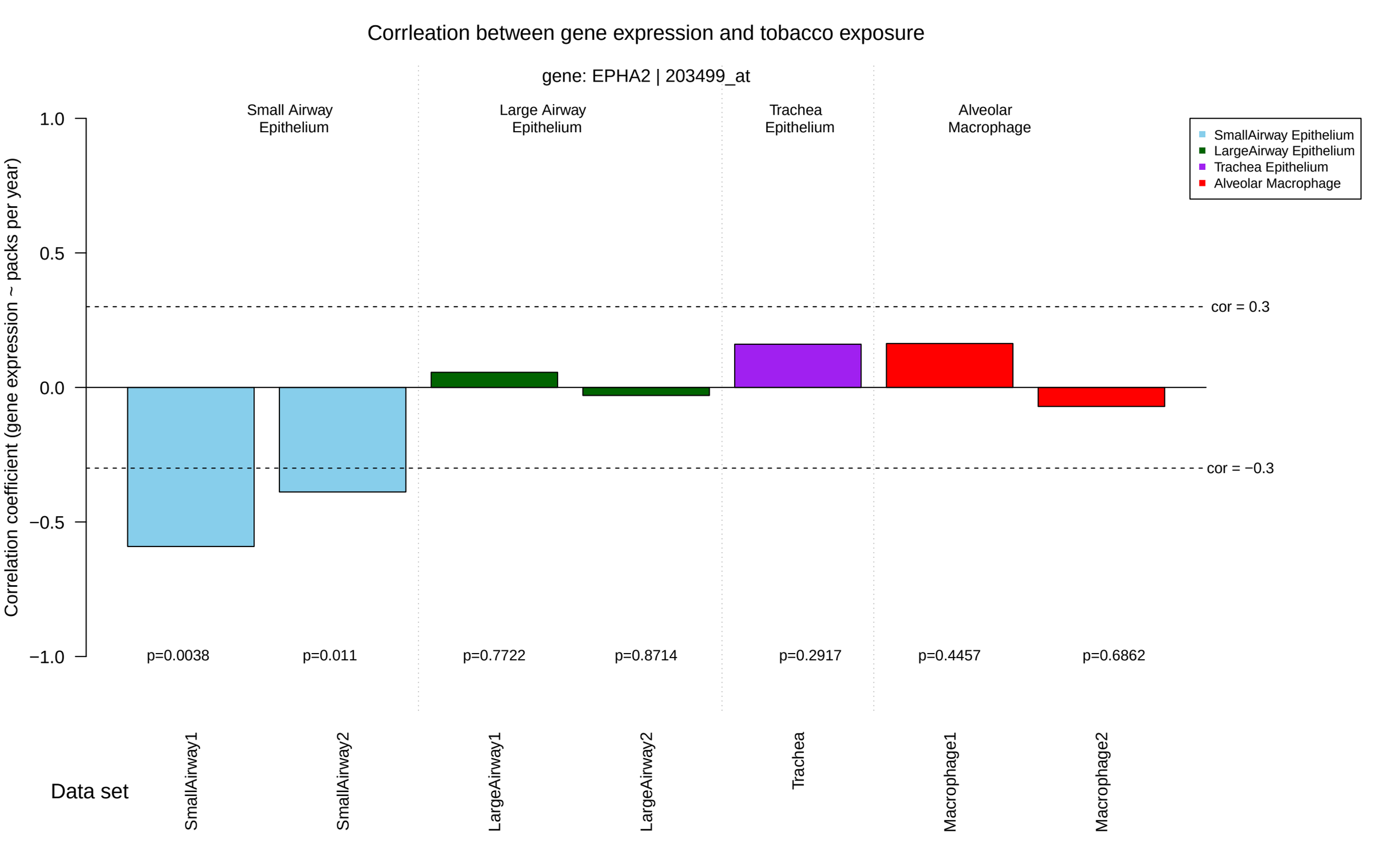

set: LargeAirway1 (p=0.7452)
EPHA2 | 203499_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9586)
EPHA2 | 203499_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1844)
EPHA2 | 203499_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0991)
EPHA2 | 203499_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9626)
EPHA2 | 203499_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EPHA2 | 203499_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0038
p=0.011
p=0.7722
p=0.8714
p=0.2917
p=0.4457
p=0.6862
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2